Table 1

Held-Out Participant 07 LOPO Classification and Transition Results Within a ±3-s Tolerance

Model	Overall		Phase-specific F1				Transition F1
	Accuracy	Macro F1	Free Motion	Approach	Painting	Retract	
Causal FSM	77.3	42.0	21.8	24.1	90.8	31.2	7.2
Random Forest	87.4	76.2	86.9	72.2	93.7	51.8	12.5
GMM	74.6	68.9	80.5	56.7	97.1	41.2	10.4
Gaussian HMM	72.8	74.5	77.2	57.9	97.0	65.8	19.3
GMM-HMM	69.6	70.0	80.4	55.2	95.1	49.3	16.5
Causal LSTM	90.4	84.4	87.1	75.8	95.7	79.0	32.5

Note. Values are percentages for held-out participant 07. Classification and phase-specific F1 scores use the previously applied same-phase ±3-s temporal-tolerance criterion. Transition F1 matches the nearest unused event with the same from-to phase direction within ±3 s. LOPO = leave-one-participant-out.